

The Kodamai Container Program: A Complete Technical Record

Part I — The Equivalence Chain (Theory)

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Abstract

This is Part I of the comprehensive, registry-grounded technical record of the Kodamai container program. It develops the theory core: the category of containers and its representation theorem, the four canonical monoidal structures on Cont/Poly , the closed structures and Day convolution, change of base over a general category, the free monad and cofree comonad, and the equivalence chain $\text{Cont} \supseteq \text{DCont} \simeq \text{Poly-comonoid} \simeq \text{Cat}$ with honest status on which links are theorems and which are conjectural. Every result carries its **trust grade in brackets**, copied verbatim from the proof registry, together with its registry node identifier. Grades are never inflated.

Grade legend. `[peer-reviewed]` > `[proved]` > `[lean-verified]` (machine-checked, scoped) / `[computed]` (finite/example evidence, not a general proof) > `[peer-claimed]`/`[in-progress]`/`[speculative]`. Appendix A (Part I) carries the full 53-node registry inventory.

Thesis of Part I. A container is a family of sets, $\text{Cont} \cong \text{Fam}(\text{Set}^{op})$; the four canonical tensors $(+, \times, \otimes, \triangleleft)$ all have the form “shapes multiply, the base’s structure appears only in positions”; and the sequential tensor $\triangleleft$ is the one whose comonoids are small categories. This is the convergence hub $\text{Cont} \supseteq \text{DCont} \simeq \text{Poly-comonoid} \simeq \text{Cat}$, developed with honest status on which links are theorems and which are conjectural.

1 The category Cont and the representation theorem

Precis. Containers $(S, P: S \rightarrow \text{Set})$, extension $\llbracket S, P \rrbracket(X) = \Sigma_s X^{P(s)}$, contravariant position maps, and $\text{Cont} \cong \text{Fam}(\text{Set}^{op})$ as the free coproduct completion of Set^{op} . Which endofunctors are containers (familial representability / wide-pullback preservation); limits and colimits; the Yoneda/representable backdrop. This is the foundation on which everything else rests.

Flag. There is **no standalone registry node** certifying the representation theorem itself; it is foundational and sourced only to the drafts `which-functors-are-containers.tex` and `category-of-containers.SEE`. This is an identified write gap (Appendix B, item 1): the “which functors are containers” result is ungraded and needs a proof file to ground it.

2 The four monoidal structures on Cont / Poly

Precis. The tensors $+ / 0$, $\times / 1$, $\otimes$ (Dirichlet) $/ y$, $\triangleleft$ (sequential) $/ y$, their units, and the Day classification; the interchange/distributivity lattice among them. $\otimes$ is the Day-ification of $\triangleleft$ via the comparator coreflection.

Theorem 2.1 (Hedges interchange table, [proved]; several cells also [lean-verified]). *All 16 cells of the 4×4 interchange table are determined, with $\times / \otimes$ corrected to non-distributing. Node `hedges-interchange-table` (2026-07-16-hedges-distributive-table.md); the cells `SeqProdDistrib`, `SeqCoproductDistrib`, `TensorCoproductDistrib`, `TimesCoproductDistrib` are additionally machine-checked.*

Proposition 2.2 (Bare $\otimes$ -monoids, [proved]). *Bare $\otimes$ -monoids = a shape monoid together with an oplax functor on fibres. Node `dirichlet-monoid-classification`.*

Proposition 2.3 (Bare $\otimes$ -comonoids, [proved]). *Bare $\otimes$ -comonoids (no $\triangleleft$) = families of monoids. Node `bare-dirichlet-comonoid`.*

Proposition 2.4 (Double comonoids, [proved]). *Double comonoids in the normal duoidal $(\text{Poly}, \triangleleft, \otimes)$ = sets of commutative monoids (an Eckmann–Hilton collapse). Node `comparator-comonoid-nogo`.*

Proposition 2.5 (Dialectica tensors, [computed]). *The two non-convolutional tensors $\ltimes / \rtimes$ are the Dialectica tensors; $\rtimes$ is left-closed. Node `other-cont-monoidal-tensors`.*

3 Closed structures and Day convolution

Precis. Which tensors are closed, the internal-hom formulas, and the obstructions. $(\text{Poly}, \times)$ is cartesian closed; Cont is CCC but not locally cartesian closed. Day convolution and its coend-free degeneration on Cont .

Theorem 3.1 (Closed Day structures, [proved]). *A Day/convolutional tensor $(\text{Cont}, \odot_*)$ is left-closed iff $(-) \star B$ is polynomial for every B , with a uniform Π -formula internal hom. Node `closed-day-structures`.*

Theorem 3.2 (Closed-tensor classification, [in-progress]). *Classification of left-closed convolutional tensors on Set : bounded arity implies $\times$ or $\vee_S$. The **infinite-arity boundary is OPEN**. Node `closed-tensor-classification` — one of the program’s two standing open fronts.*

Proposition 3.3 (Dirichlet internal hom, [lean-verified]). *$(\text{Cont}, \otimes, y)$ is closed with the Dirichlet internal hom (`DirichletClosed.lean`, `DirichletHomPi.lean`). The mathematics is Niu–Spivak prior art; the proof object is ours.*

4 Change of base — containers over a general category

Precis. $\text{Cont}_C = \text{Fam}(C^{op})$; the fullness criterion; closedness criteria T2/T4; the fibrational reading; M-containers; containers over Vec .

Theorem 4.1 (T1 (flagship), [proved]). *$\llbracket - \rrbracket : \text{Fam}(C^{op}) \rightarrow [C, C]$ is full-and-faithful iff the monoidal unit I is connected (i.e. $C(I, -)$ preserves coproducts) — not extensivity. Node `fullness-unit-connected`.*

Theorem 4.2 (T2, [proved]). *Dirichlet/Day closedness over a general base = familial representability. Node `t2-day-closedness-famcop`.*

Theorem 4.3 (T4-left, [proved]). $\triangleleft$ left-closedness on $\text{Fam}(C^{op})$. Node `t4-left-closedness-lhd-famcop`.

Theorem 4.4 (M-containers, [peer-reviewed]). $M\text{-Cont} = \text{Fam}(\text{Kl}(M)^{op})$; the extension is faithful always, full iff M is codense and connected; the three enriched extensions collapse to $\{\text{Id}\}$. Rick-refereed. Node `change-of-base`.

Theorem 4.5 (M-container trilogy, [proved]). $\text{THM1}/\text{THM2}/\text{THM3}$ (composition $\iff$ polynomiality), with a necessity trichotomy. Node `m-containers`.

Proposition 4.6 (Logic of containers, [proved]). $\text{Cont}(\text{cod})$ is the bifibration $\text{Fam}(\text{Set}^{op})$, with shape-level Fam-Kan quantifiers. Nodes `cont-cod-predicate-fibration`, `joint-bc-cont-cod`.

Proposition 4.7 (Containers over Vec, [proved]). $\text{LinCont} = \text{Fam}(\text{Vec}^{op})$, with biproduct collapse; the extension is faithful but **not** full over Vec. Node `linear-containers-vec`.

Flag. Correction (MEMORY): the earlier VCont “faithful” claim over Vec was wrong — use the corrected statement (faithful-but-not-full) above.

Proposition 4.8 (Left-adjoint behaviour over Vec, [proved]). $(-)\triangleleft q$ has the predicted left-adjoint behaviour over $\text{Fam}(\text{Vec}^{op})$, but summability does not gate it as predicted (node `left-adjoint-over-vec`); $L_q = (-)\triangleleft q$ is a parametric right adjoint (node `pra-vs-probe-method`); the copowers/writer-monad gap (Q) reformulation holds in ZFC (node `copowers-gap-writer-monad`, [proved, ZFC]).

Proposition 4.9 (Fibredness vs. left-closure, [proved]). The $\text{BHM} \triangleright$ left-variable fibredness result. Node `fibredness-vs-left-closure`.

5 Free monad and cofree comonad in Poly

Precis. The free monad C^* (well-founded trees, grafting) and cofree comonad C^∞ (M-type, node-paths) of a container, and the linear/tensor-algebra instance.

Theorem 5.1 (Free monad, [lean-verified]). The free monad = the $\triangleleft$ -monoid $(S^* \triangleleft P^*, \text{graft}, \text{lf})$; the three monoid laws are machine-checked (*Quot.sound-only*). Node `free-monad-grafting` (*Free.lean*).

Theorem 5.2 (Cofree comonad, [proved]). The cofree comonad C^∞ is the $\triangleleft$ -comonoid on the M-type of p -trees; counit = root, comultiplication = subtree-relabel + path-concat. Node `cofree-comonad`.

Theorem 5.3 (O’Neill linear instance, [proved]). O’Neill’s free monad on linear self-attention = the free $\triangleleft$ -monoid on a one-shape linear container = the tensor algebra $T(A)$; with Adámek chain / residual-as-pointing. Node `oneill-free-monad-linear-container`.

6 The equivalence chain and its honest status

Precis. $\text{Cont} \supseteq \text{DCont} \simeq \text{Poly-comonoid} \simeq \text{Cat}$; the directed-container laws D1–D5; the Lean milestone ladder M1–M6; what is proved vs. conjectural.

Theorem 6.1 (Equivalence chain M1–M6, [in-progress]). Node `equivalence-chain`. **M1–M4 are done and partly Lean-verified** (container library, comonad/directed-container equivalence, ZS characterisation, $\text{DCont} \simeq \text{Cat}$ — see Part V); **M5 (bridge to Poly, four monoidal structures) = Clarke–Di Meglio, and M6 (double category Org, dynamic categories) are OPEN / conjectural**. M5/M6 must not be stated as proved.

Remark. $\text{Cat}^\# = \text{DCont} \cong \text{Cof}$ in Poly; $\text{DCont} \simeq \text{Cat}$ is the convergence hub (source `convergence-hub.tex`, connection `dcont-cat-is-the-convergence-hub`).

A Full registry inventory (53 nodes)

Grade legend: PR = peer-reviewed, P = proved, LV = lean-verified, C = computed, IP = in-progress, PC = peer-claimed. Grade breakdown: **3 peer-reviewed, 38 proved, 4 lean-verified, 5 computed, 2 in-progress, 1 peer-claimed, 0 explicitly speculative.**

#	registry file	grade	one-line gist
1	change-of-base	PR	M-container extension f&f classification; 3 enriched exts collapse to {Id}
2	holonomy-composition-zs-bridge	PR	update monads sharing S compose as ZS, welding holonomy
3	tangent-uniqueness-spoly	PR	∂ is the unique nontrivial representable tangent on SPoly (corrected stmt)
4	free-monad-grafting	LV	free monad = $\triangleleft$ -monoid; 3 monoid laws machine-checked
5	cyclic-h2-norm-engine	LV	$H^2(C_n; \mathbb{Z}) = \mathbb{Z}/n$ via norm map
6	cyclic-h2-gcd-engine	LV	$H^2(C_n; \mathbb{Z}/m) = \mathbb{Z}/\gcd(n, m)$, unified engine
7	fibre-h2-z2-engine	LV	$H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$, nonsplit- $\mathbb{Z}/4$ vs Klein witnesses
8	equivalence-chain	IP	DCont=Poly-comonad=Cat; M1-M4 done, M5/M6 OPEN
9	closed-tensor-classification	IP	left-closed convolutional tensors; infinitarity OPEN
10	clio-peer-claims	PC	Clio's asserted results (transfer operators etc.)
11	game-comonad-poly	C	EF/pebble/modal game comonads factor through Poly
12	groupoid-zs-obstruction	C	groupoid ZS merge $\iff [\omega] = 0$; $cd \leq 1 \implies$ merge
13	other-cont-monoidal-tensors	C	$\ltimes/\rtimes$ are the Dialectica tensors; $\ltimes$ left-closed
14	quiver-skew-brace-zs	C	Ferri prunability = ZS global-obstruction vanishing; = descent
15	supply-chain-zs	C	supply chain = DCont; shared-warehouse composition = ZS
16	bare-dirichlet-comonoid	P	bare $\otimes$ -comonoids = families of monoids
17	cartan-scalar-object-spoly	P	SPoly tangent bundle = monoids, no negation $\implies$ Cartan fails
18	closed-day-structures	P	Day tensor left-closed iff $(-) \star B$ polynomial; Π internal hom
19	cofree-comonad	P	cofree comonad = $\triangleleft$ -comonoid on M-type of p -trees
20	comparator-comonoid-nogo	P	double comonoids in $(\text{Poly}, \triangleleft, \otimes)$ = sets of commutative monoids
21	composition-nerve-level	P	grade by nerve level; composition = $N_2 = \text{degree } 1$; $N_2 \subsetneq N_3$ in assembly
22	cont-cod-predicate-fibration	P	$\text{Cont}(\text{cod}) = \text{Fam}(\text{Set}^{op})$ bifibration
23	copowers-gap-writer-monad	P	(Q) copowers/writer-monad reformulation (ZFC)
24	dirichlet-monoid-classification	P	bare $\otimes$ -monoids = shape monoid + oplax functor on fibres
25	effect-coeffect-arrows	P	effect/coeffect arrow face iff non-branching; class $E + A \times X$
26	emergent-holonomy-is-ext-tower	P	meeting count = $\text{Ext}_{kG}^n = \oplus H^n$, the Ext tower
27	fibredness-vs-left-closure	P	BHM $\triangleright$ left-variable fibredness result
28	fullness-unit-connectedness	P	T1: $\llbracket - \rrbracket$ f&f $\iff$ unit connected (NOT extensivity)
29	holonomy-triviality	P	liftings $\cong$ -Cat collapse engine
30	joint-bc-cont-cod	P	shape-level Fam-Kan quantifiers over $\text{Cont}(\text{Set}) = \text{Poly}$
31	left-adjoint-over-vec	P 5	$(-) \triangleleft q$ left adjoint over $\text{Fam}(\text{Vec}^{op})$
32	lie-parallelizable-poly-comonoid	P	poly comonoid parallelizable $\iff$ out-degree; Lanfranchi port ill-posed
33	linear-containers-vec	P	LinCont = $\text{Fam}(\text{Vec}^{op})$; faithful-not-full

B Identified write gaps

These are honestly-flagged gaps in the current write-up (not mathematical gaps unless stated):

1. **Representation theorem has no registry node** (§1): the foundational “which functors are containers” is sourced only to `.tex`, ungraded. Needs a proof file.
2. **Inventory $H^1[\theta_R]$ has no standalone node** (see Part IV): it lives inside `composition-nerve-level`; needs its own registry entry if cited as an independent degree-one result.
3. **Tax computation (SEED Path 5) has no write-up and no node** (see Part IV): either model it or mark as explicit future work.
4. **GA empirical result is external** (see Part IV): Langer–Vega GECCO / ga-containers ACT are collaborator publications, not MacBeth nodes; cite as such.
5. **The nerve-note N2 correction (2026-10-09) must be propagated**: the demoted weld- ω -as-non-associativity gloss must be restated as a *splitting* obstruction throughout (Parts II, IV), and “nerve level” disambiguated.
6. **Day-convolution treatment exists but is scattered** (§3): consolidation, not fresh proving.
7. **Effect “zoo” / update monad**: the Ahman–Uustalu update monad has no MacBeth write-up of its own; the biggest *synthesis* gap, though the component results (`monad-comonad-entwining`, `state-object-delta`, `effect-coeffect-arrows`) are all **[proved]**.